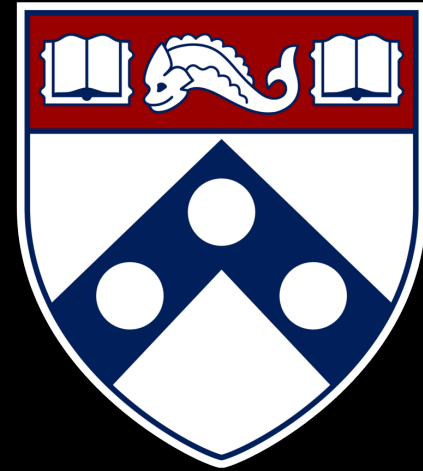


Empirical Security & Privacy, for Humans

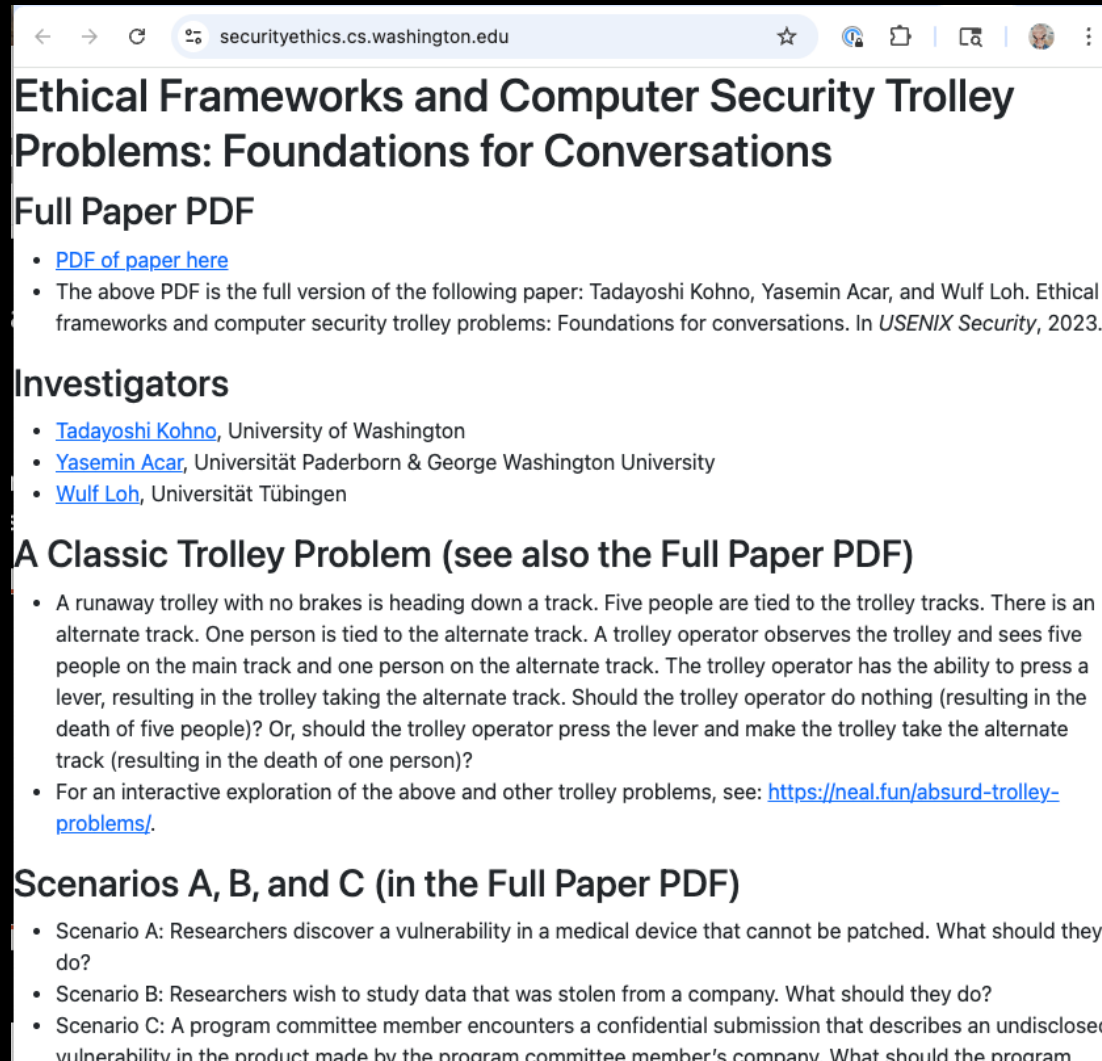


UPenn CIS 7000-010
10/14/2025



Ethics in security research;
Password managers & security

Readings



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Ethical Frameworks and Computer Security Trolley Problems: Foundations for Conversations

Full Paper PDF

- [PDF of paper here](#)
- The above PDF is the full version of the following paper: Tadayoshi Kohno, Yasemin Acar, and Wulf Loh. Ethical frameworks and computer security trolley problems: Foundations for conversations. In *USENIX Security*, 2023.

Investigators

- [Tadayoshi Kohno](#), University of Washington
- [Yasemin Acar](#), Universität Paderborn & George Washington University
- [Wulf Loh](#), Universität Tübingen

A Classic Trolley Problem (see also the Full Paper PDF)

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Scenarios A, B, and C (in the Full Paper PDF)

- Scenario A: Researchers discover a vulnerability in a medical device that cannot be patched. What should they do?
- Scenario B: Researchers wish to study data that was stolen from a company. What should they do?
- Scenario C: A program committee member encounters a confidential submission that describes an undisclosed vulnerability in the product made by the program committee member's company. What should the program



Better managed than memorized?

Studying the Impact of Managers on Password Strength and Reuse

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Abstract

Despite their well-known security problems, passwords are still the incumbent authentication method for virtually all online services. To remedy the situation, users are very often referred to password managers as a solution to the password reuse and weakness problems. However, to date the actual impact of password managers on password strength and reuse has not been studied systematically.

We provide the first large-scale study of the password managers' influence on users' real-life passwords. By combining qualitative data on users' password creation and management strategies, collected from 476 participants of an online survey, with quantitative data (incl. password metrics and entry methods) collected in situ with a browser plugin from 170 users, we were able to gain a more complete picture of the factors that influence our participants' password strength and reuse. Our approach allows us to quantify for the first time that password managers indeed influence the password security, however, whether this influence is beneficial or aggravating existing problems depends on the users' strategies and how well the manager supports the users' password management right from the time of password creation. Given our results, we think research should further investigate how managers can better support users' password strategies in order to improve password security as well as stop aggravating the existing problems.

1 Introduction

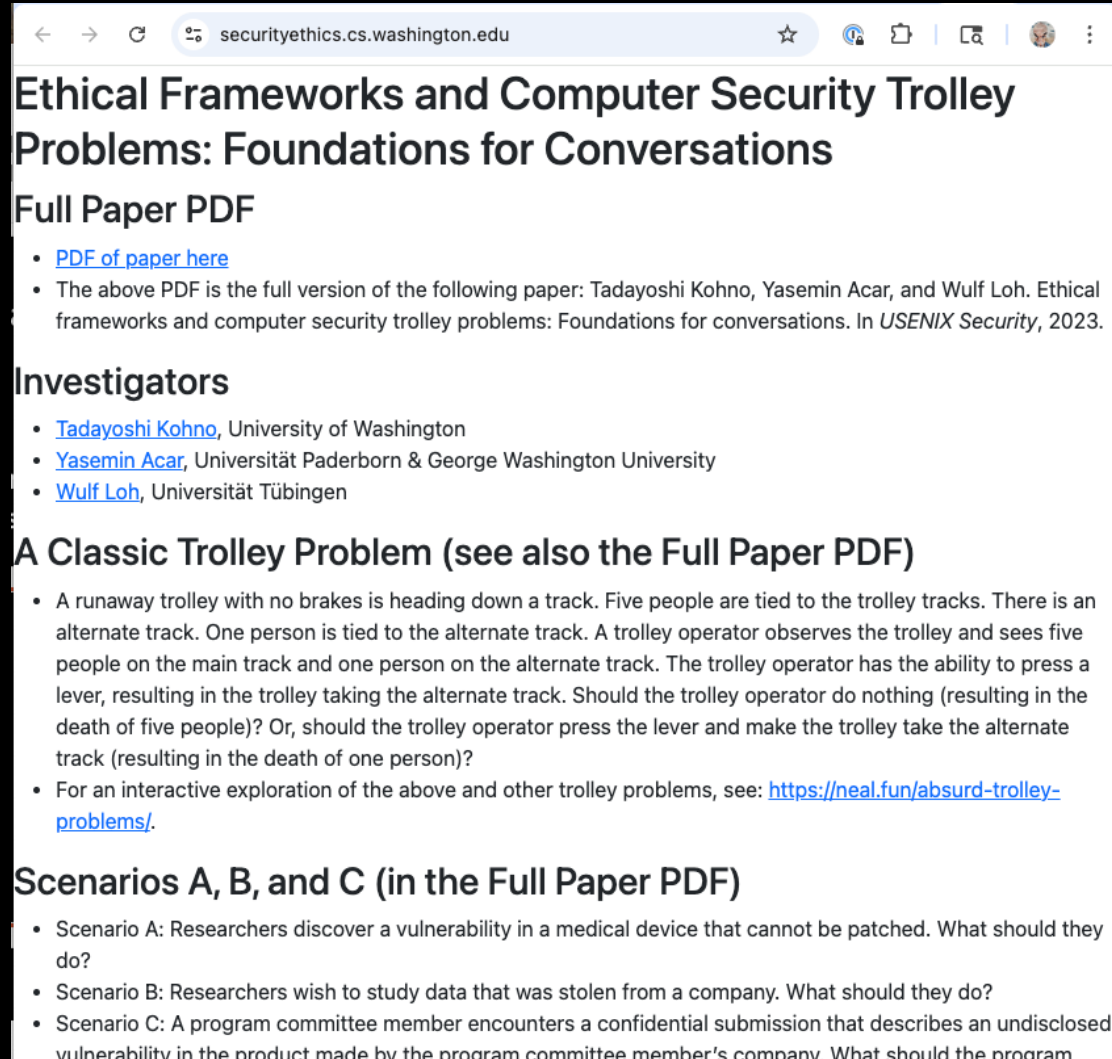
For several decades passwords prevail as the default authentication scheme for virtually all online services [44, 11, 30]. At the same time, research has again and again

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In the past, different solutions have been implemented to help users creating stronger passwords, such as password meters and policies, which are also still subject of active research [41, 54, 17, 45, 68]. Among the most often recommended solutions [28, 59, 53, 62, 56] to these problems for end-users is technical support in the form of password management software. Those password managers come built-in to our browsers, as a browser plugin, or as separate applications. Password managers are being recommended as a solution because they fulfill important usability and security aspects at the same time: They store all the users' passwords so the users do not have to memorize them; they can also help users entering their passwords by automatically filling them into log-in forms; and they can also offer help in creating unique, random passwords. By today, there are several examples of third party password managers that fit this description, such as Lastpass [5], 1Password [1], and even seemingly unrelated security software, such as anti-virus [4] solutions.

Unfortunately, it has not been sufficiently studied in the past whether password managers fulfill their promise and indeed have a positive influence on password security or not? To break this question down, we are interested in 1) whether password managers actually store strong passwords that are likely auto-generated by, for instance, password generators, or if they really are just storage where users save their self-made, likely weak passwords? Further, we are interested whether 2) users, despite using password managers, still reuse passwords across different websites or if do they use the managers' support to main-

Readings



The screenshot shows a web browser window with the address bar displaying 'securityethics.cs.washington.edu'. The page title is 'Ethical Frameworks and Computer Security Trolley Problems: Foundations for Conversations'. Below the title, there is a section 'Full Paper PDF' with a link 'PDF of paper here' and a paragraph stating that the PDF is the full version of a paper by Tadayoshi Kohno, Yasemin Acar, and Wulf Loh, published in 'USENIX Security, 2023'. The next section is 'Investigators', listing three authors with their affiliations: Tadayoshi Kohno (University of Washington), Yasemin Acar (Universität Paderborn & George Washington University), and Wulf Loh (Universität Tübingen). The following section is 'A Classic Trolley Problem (see also the Full Paper PDF)', which describes a trolley problem scenario and provides a link to an interactive exploration. The final section is 'Scenarios A, B, and C (in the Full Paper PDF)', listing three scenarios: Scenario A (researchers discover a vulnerability in a medical device), Scenario B (researchers wish to study stolen data), and Scenario C (a program committee member encounters a confidential submission).

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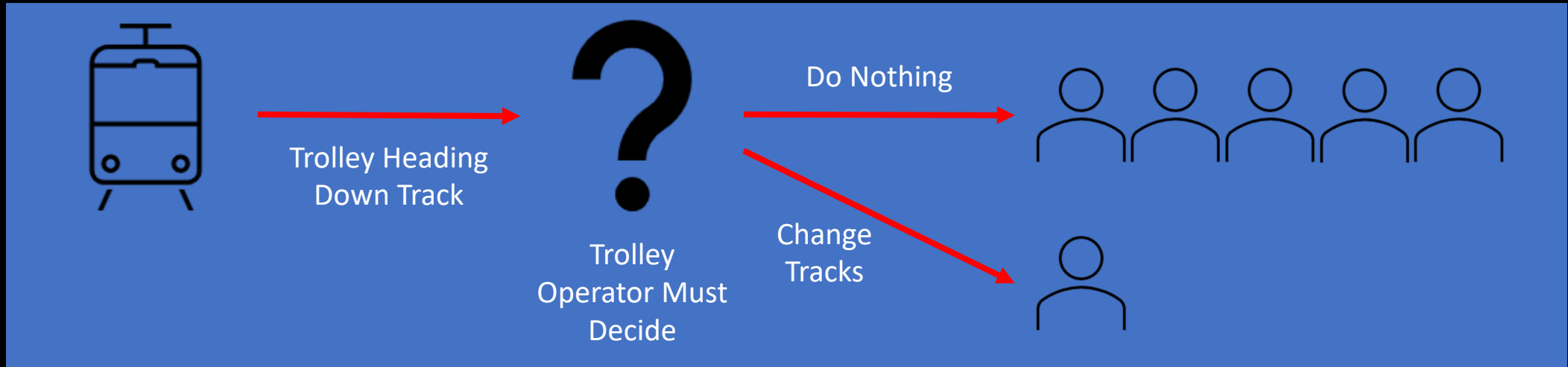
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- A runaway trolley with no brakes is heading down a track. Five people are tied to the trolley tracks. There is an alternate track. One person is tied to the alternate track. A trolley operator observes the trolley and sees five people on the main track and one person on the alternate track. The trolley operator has the ability to press a lever, resulting in the trolley taking the alternate track. Should the trolley operator do nothing (resulting in the death of five people)? Or, should the trolley operator press the lever and make the trolley take the alternate track (resulting in the death of one person)?
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A Classic Dilemma: The Trolley Problem



The Trolley Problem is a classic thought experiment / ethical dilemma (Philippa Foot).

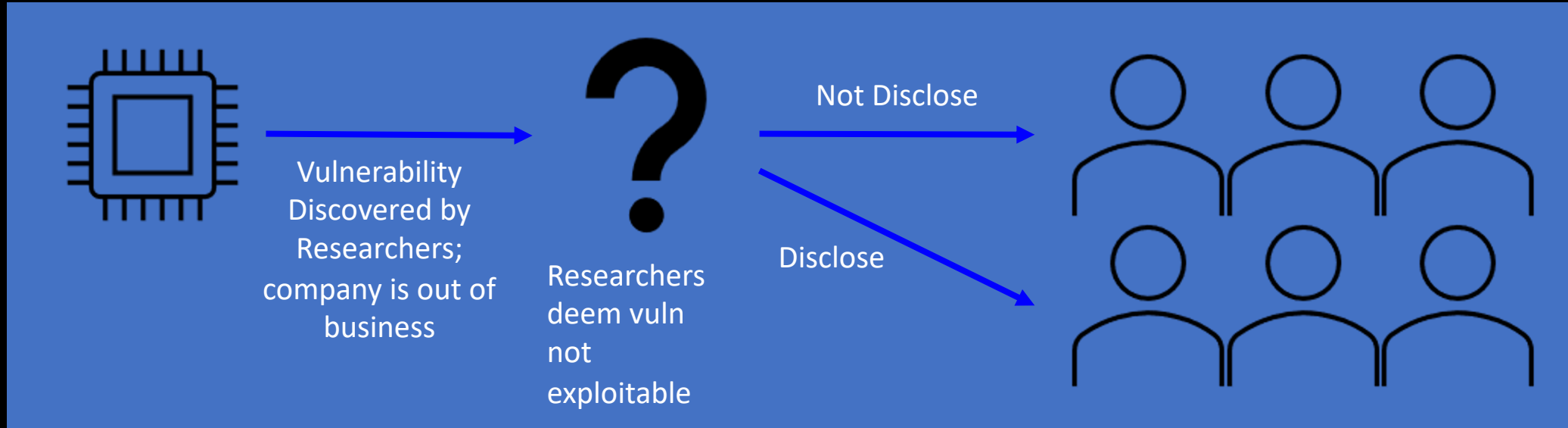
- A runaway trolley with no brakes is heading straight. **Five people** are tied to those tracks. **One person** is tied to an alternate set of tracks.
- Should the **operator do nothing** (five people die) or **change the path** of the trolley (one person dies)?

These slides based on slides by the paper's authors; used with permission

<https://securityethics.cs.washington.edu/>

Scenario A: Medical Device Vulnerability

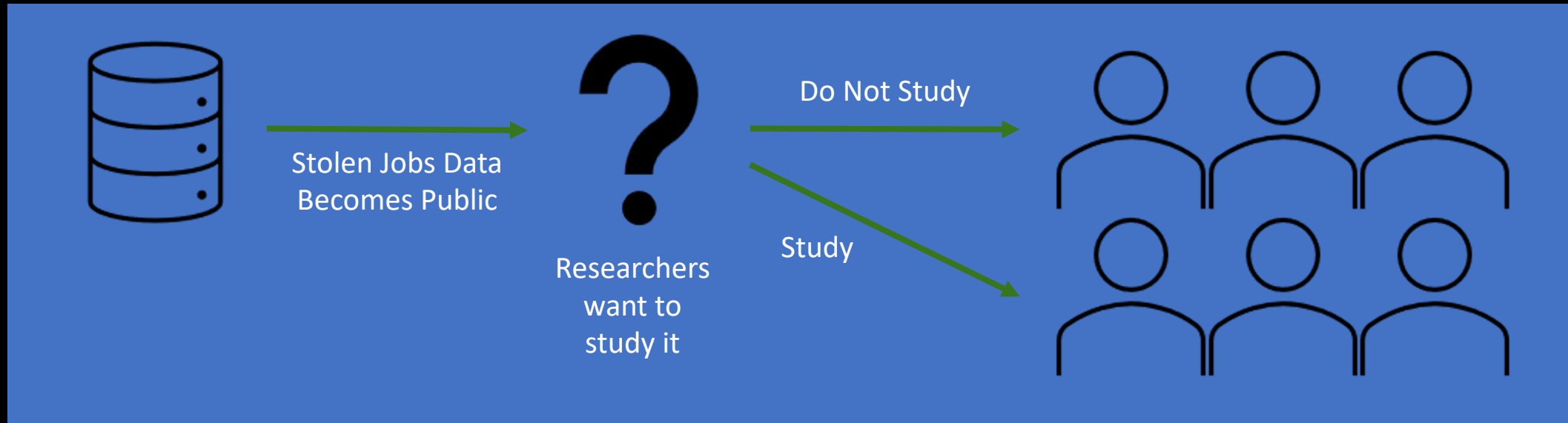
(some details may not reflect reality)



- **If not disclosed:** Patients have **no awareness** that their device is vulnerable; **patients** keep and/or proceed with obtaining device and **receive** significant **health benefits**
- **If disclosed:** Patients have the **choice** to not receive or to remove the device; **risk** of **psychological harm** if **patients know** they have a **vulnerable device** (even if chance of exploitation is zero); **risk** of **health harm** if patients do not receive / remove the device

Scenario B: Immorally-Obtained Data

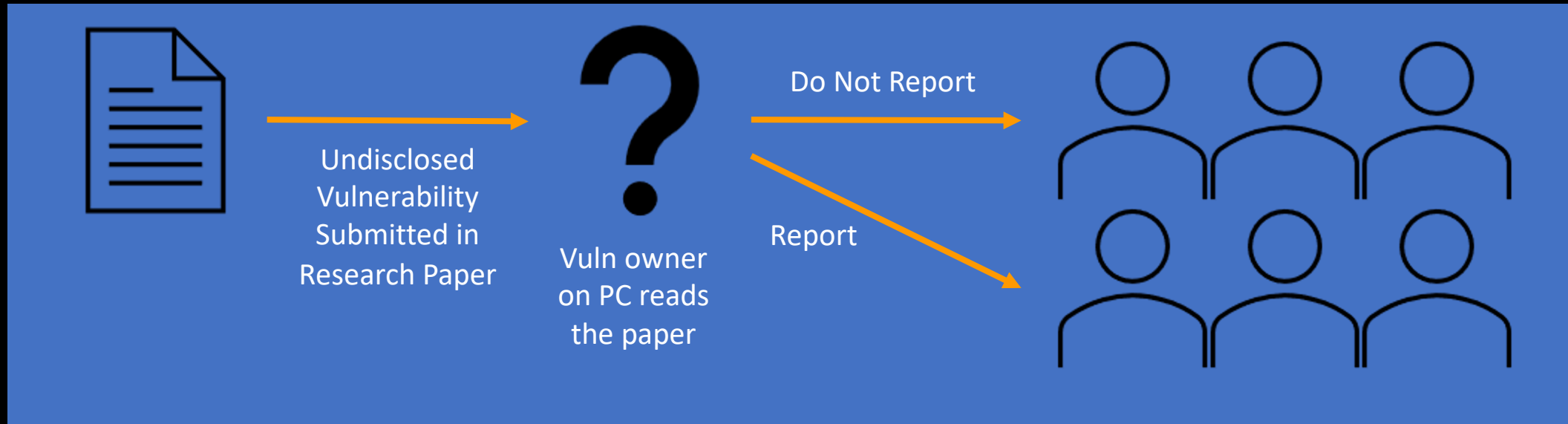
(some details may not reflect reality)



- **If not studied:** The stolen data is from people; in choosing this option, the researchers would not have further violated their privacy and their right to informed consent
- **If studied:** Result will help prevent future job applicants from being harmed by biased or a vulnerable job-to-applicant AI matching system

Scenario C: Inadvertent “Disclosure”

(some details may not reflect reality)



- **If not reported:** Confidentiality of PC process preserved; authors have agency over when and how they disclose
- **If reported:** Company C checks to see if the vulnerability is already being exploited; Company C can immediately begin working on solutions to protect their users; users' security, privacy, and safety will be protected before the vulnerability becomes public

How to evaluate these choices?

Review them using **moral / ethical frameworks**

- **Ethics**: The academic / professional engagement with moral questions
- **Morality**: The scope of this engagement, i.e., the sum of all moral considerations, what we owe each other, e.g., not to harm others, respect their rights
 - E.g., use **ethics** to determine if something (e.g., an action) is **moral**

Consequentialist Ethics (Utilitarianism)

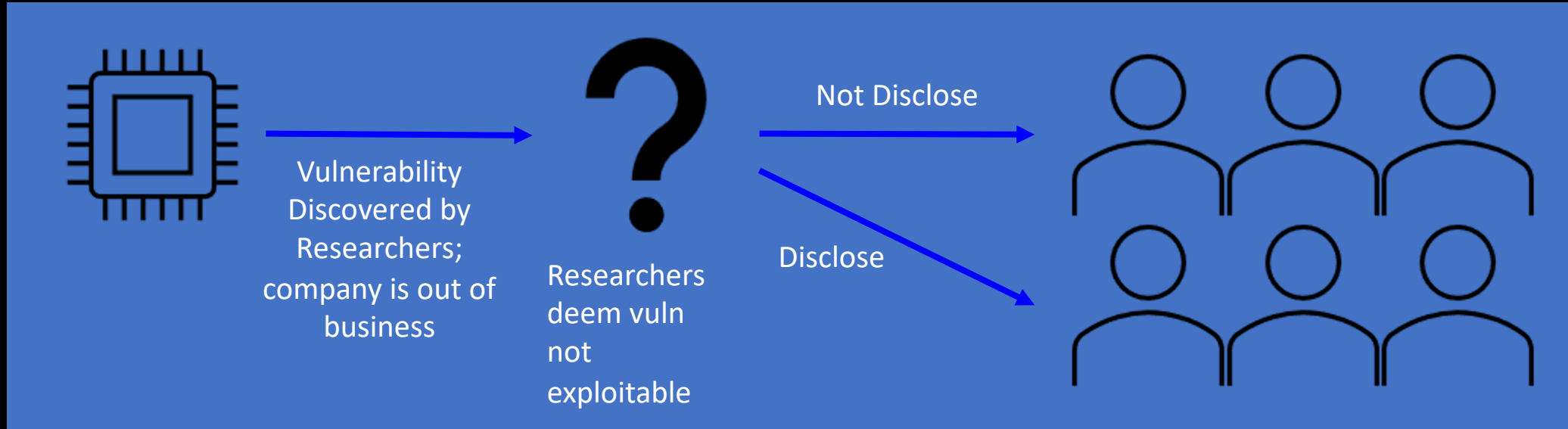
- Focuses on **consequences** of actions, policies, institutions
- **Utilitarianism**: example of consequentialism in which consequences are measured with respect to well-being
- Different instances of utilitarianism take different definitions for well-being:
 - **Hedonic utilitarianism**: Actions are morally right if they promote the **greatest happiness** for the **greatest number of people**
 - **Objective list utilitarianism**: Actions are morally right if they promote the **greatest benefits** from an **objective list** for the **greatest number of people** (e.g., **health, wealth, access to resources**)
- To assess, **count numbers** and **weigh benefits / harms**, e.g., five deaths are worse than one → **consequentialists would change the trolley's tracks**

Deontological Ethics (Fundamental Rights)

- **Deontological** ethics focuses on the moral **duties** of a moral **actor**, e.g., an individual or institution
- **People have fundamental rights**; moral actors have a **duty** to **respect** those rights
 - Example rights: The **right** to **privacy**, the right to **self-agency**, the right to **informed consent**
- **Kantian deontological ethics**: One should **not violate any single person's rights in order to accomplish another objective**
 - Kant in describing this "Categorical Imperative": *human beings should be treated as "ends and never purely as means"*
 - **Changing the trolley tracks would violate one person's right** (their right to live) **in order to accomplish** the **saving of five other lives**; changing the track would **use that single person as an "means", not as an "ends"**; **under Kantian deontological ethics → do not change the trolley's tracks**

Scenario A: Medical Device Vulnerability

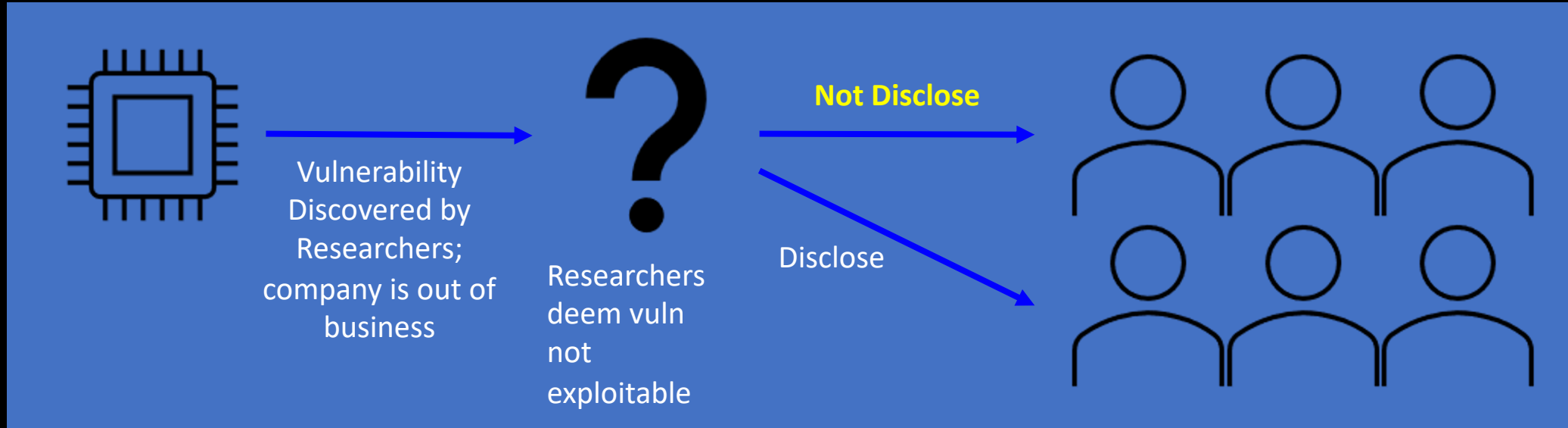
(some details may not reflect reality)



Researchers find a **vulnerability** in a **medical device**; device manufacturer is out of business. Should the researcher disclose the vulnerability to doctors, patients, and the public? Should the researchers keep the vulnerability secret?

Scenario A: Medical Device Vulnerability

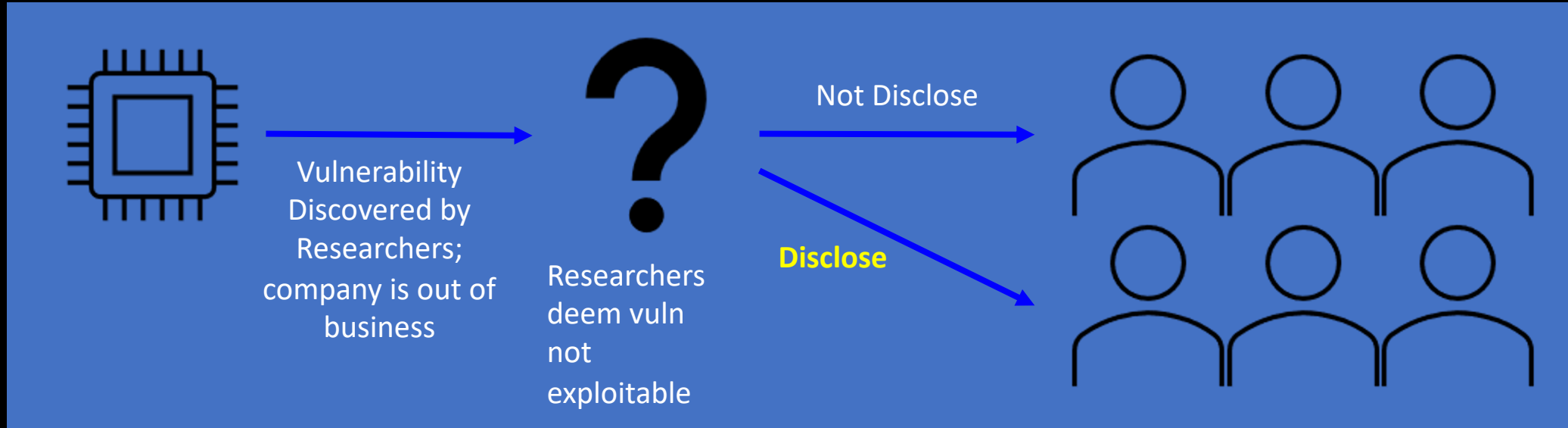
(some details may not reflect reality)



Consequentialist Ethics: **Likelihood of exploit is zero; harms if patients informed**
(health: remove device / not get device; happiness: live with knowledge that the device has faults) → **do not disclose vulnerability**

Scenario A: Medical Device Vulnerability

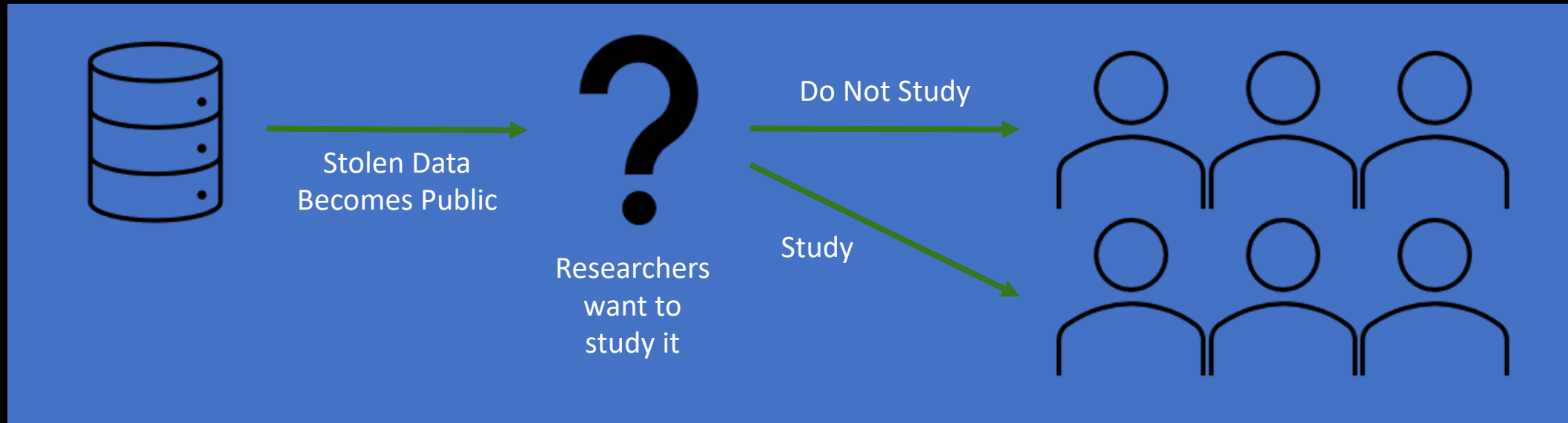
(some details may not reflect reality)



Deontological Ethics: Duty to respect people's **right to informed consent** (e.g., warnings on medicine ads) and **right to self-agency** (make their own decisions about what is best for them) → **disclose vulnerability**

Scenario B: Immorally-Obtained Data

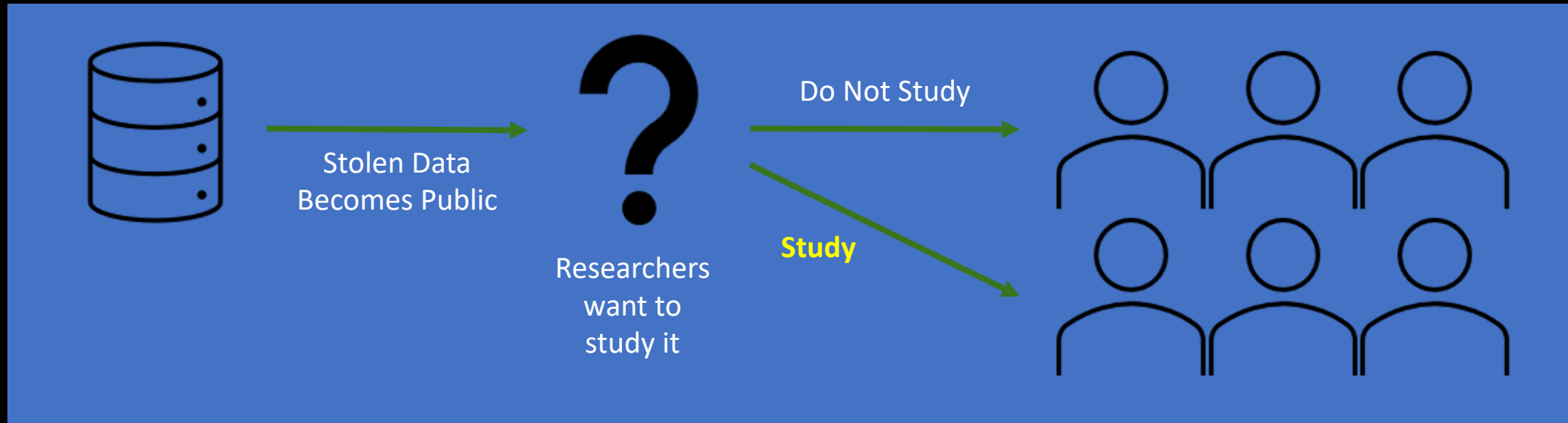
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Adversaries **stole data** from a job-to-applicant matching system, which includes significant private information about job applicants. Should researchers study that data if doing so can benefit future job applications (e.g., decrease future racial and gender biases and the risk of manipulation in the job-to-applicant matching AI)?

Scenario B: Immorally-Obtained Data

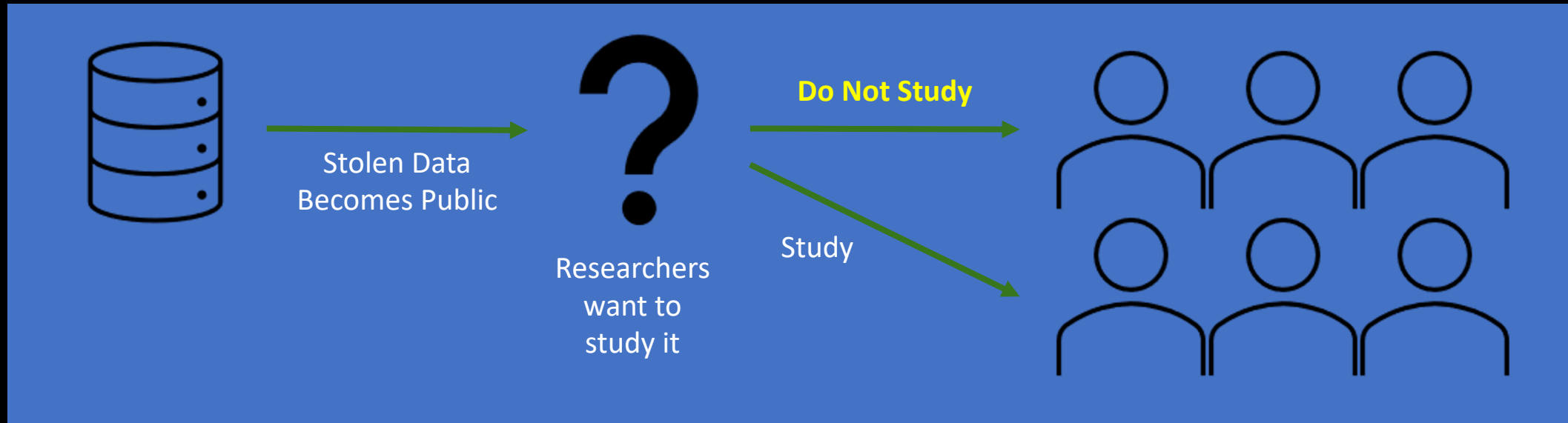
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Consequentialist Ethics: **Benefits** of studying the data to society and to future job applicants is **high**; the data was already made “public”, which itself caused harms, but the study of the data does not create new harms → **study the data**

Scenario B: Immorally-Obtained Data

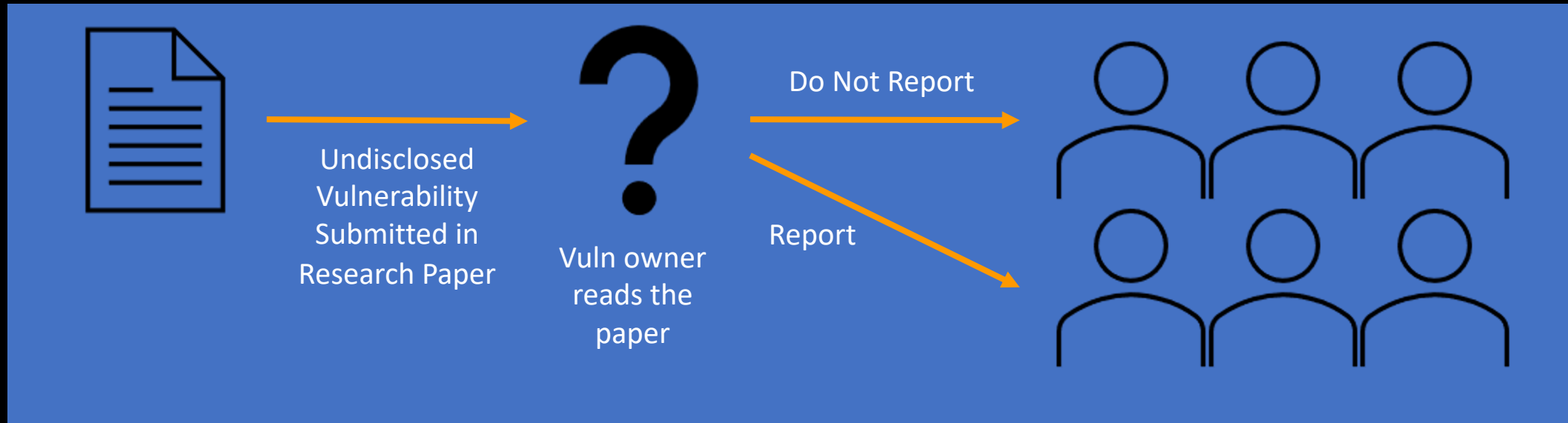
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Deontological Ethics: People have a **right to privacy** and **informed consent**; rights extend to data **intended** to be private even if researchers have access; impacted **job applicants explicitly requested the data to be deleted**; while studying data could improve the rights of future job applicants, doing so would be **at the expense of the rights of those impacted by the data breach** → **do not study the data**

Scenario C: Inadvertent “Disclosure”

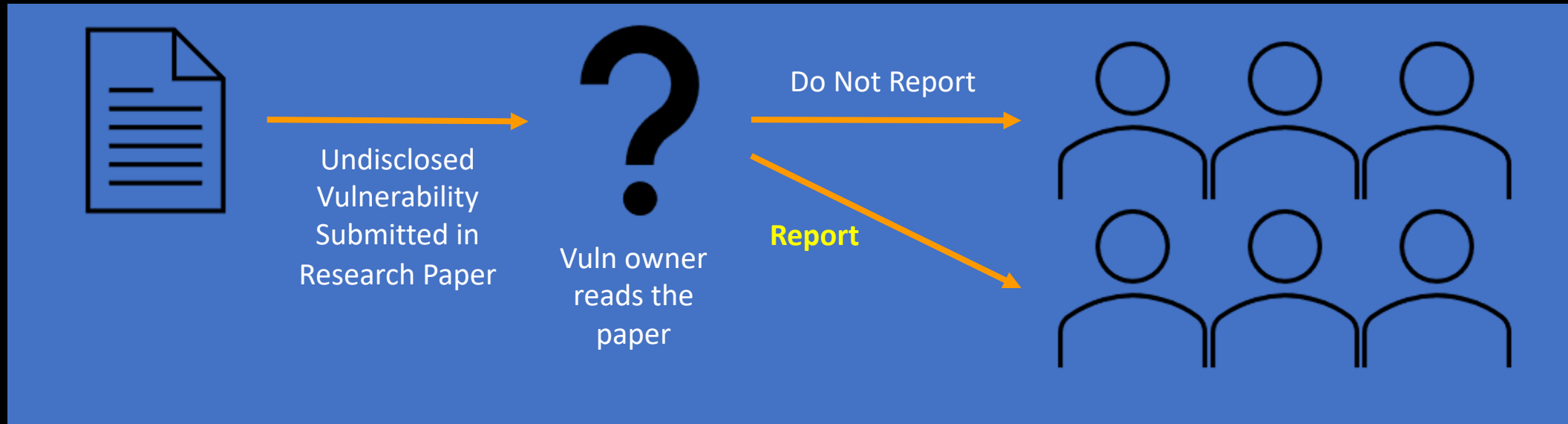
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Researchers submit a paper with an undisclosed vulnerability in the product from Company C to a peer-reviewed conference. An **employee at Company C is on the conference program committee**. Should the employee disclose the vulnerability to their company?

Scenario C: Inadvertent “Disclosure”

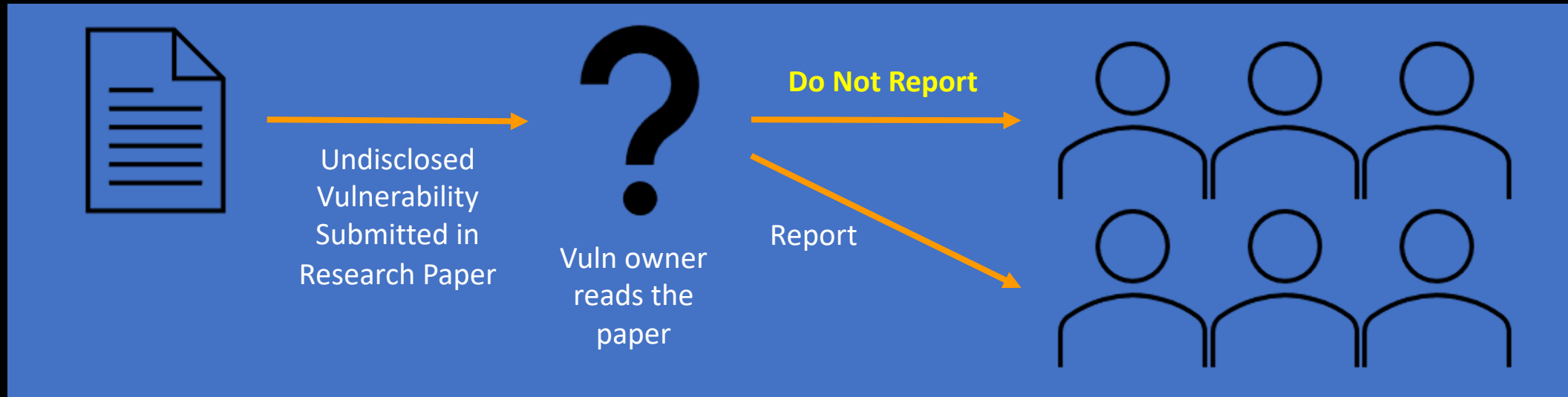
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Consequentialist Ethics: Possible **harms to authors** if employee of Company C **discloses vulnerability** to Company C; significant **benefits** to Company C’s **users** if the employee **discloses vulnerability** to Company C; the **benefits / harms** must be weighed → Employee of Company C **should report** the vulnerability to Company C

Scenario C: Inadvertent “Disclosure”

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Deontological Ethics: The program committee member may feel a sense of **duty** to **Company C** and **their users**; program committee members also have **duty** to respect the **autonomy** of **authors** and the **confidentiality** of the **peer review process**; the employee **agreed** to **respect this duty** when they joined the **program committee**, as did Company C’s leadership team; if **breaking** the **confidentiality** of the peer review process is allowed, the **whole institution of the peer review process would not be possible** → Company C employee **should not report** the vulnerability to Company C

Discussion

- What do you think about the scenarios, and how they were made?
- How would you approach these situations? What would you decide?
- How has reading about these problems made you think about ethical choices or judgments you might make
 - as a researcher / paper author
 - as a research reviewer / committee member
 - as a reader
- What are the limitations of what's presented?
- Is this methodology useful in other fields?

Their recommendations

- Researchers: Ethics modeling, to help decisionmaking
 - Identify stakeholders, identify assets, specifically identify benefit/harm, rights/duties.
- PC members: Not only whether right/wrong, but why
- Community: How to build supports for good decisions

Readings

Better managed than memorized? Studying the Impact of Managers on Password Strength and Reuse

Sanam Ghorbani Lyastani
CISPA, Saarland University

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1 Introduction

For several decades passwords prevail as the default authentication scheme for virtually all online services [44, 11, 30]. At the same time, research has again and again

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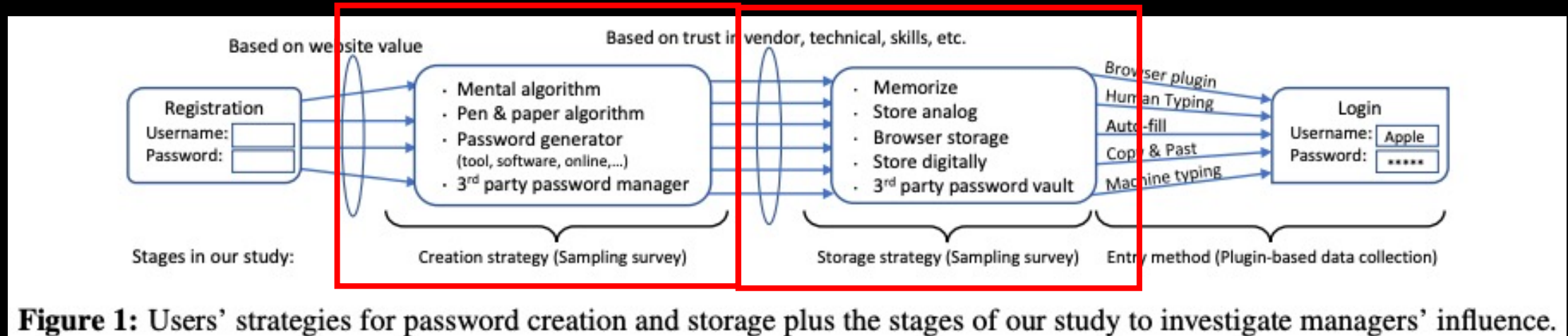
Activities of password management

- Password **creation** – with algorithm, by hand, using service, ...
- Password **storage** – memorized, in file, in service, ...
- Password **entry** – typed, cut&paste, auto-filled, ...
- Password managers can handle some/all of these activities, potentially improving security
 - Creating passwords that are difficult to guess (and hard to remember)
 - Reducing password reuse

Impact of password managers on security

- RQ: Impact of password managers on password strength and reuse?
- Data collected: Survey on Mechanical Turk, then data collection from some of these about password management using a Chrome plugin
 - Plugin used for four days
- Data analysis: Correlate factors against measures of password strength and occurrences of reuse
 - Factors include how exactly password managers were used or not used
 - Allows relating to baselines, identifying key aspects of success

Activities of password management



- Password **creation** – with algorithm, by hand, using service, ...
- Password **storage** – memorized, in file, in service, ...
- Password **entry** – typed, cut&paste, auto-filled, ...

Survey questions

- 31-34 questions, in six groups
 - general privacy attitude,
 - their attitude towards passwords,
 - their skills and strategies for creating and managing passwords, including
 - an assessment of the strength of a sample of passwords
 - whether they admit to reusing passwords
 - which password manager they use (if any)
 - basic demographic questions
 - attention check questions
- Qualitative answers independently coded

Plugin implementation: Measuring effects

- To determine password strength, measured:
 - Shannon entropy
 - NIST entropy
 - zxcvbn score, ranges 0 to 4 – intends to correlate with “crackability”
- To determine (partial) reuse: Stored keyed hashes of passwords
 - Key stored locally
 - Hash on full password and on every 4-character substring

Plugin implementation: Measuring factors

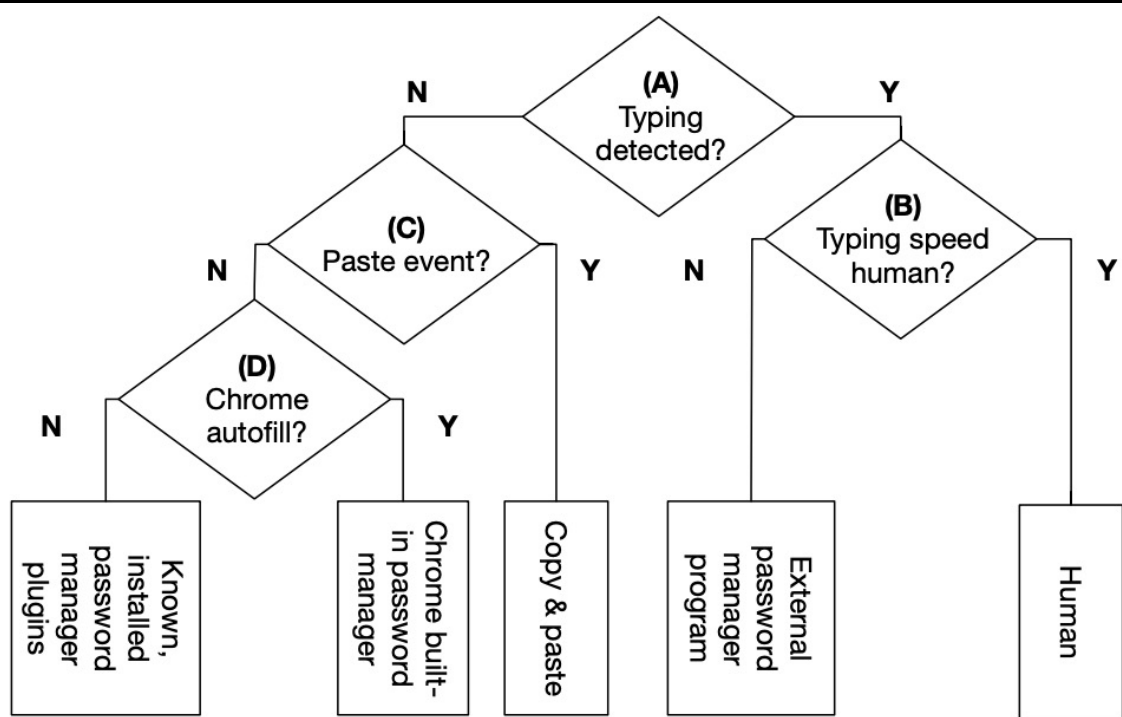


Figure 2: Decision tree to detect password entry methods

Note: You will see this pop-up for this URL (twitter.com) only once. Please answer all questions properly.

Privacy Protection Note: We never store your password anywhere! If you are interested in which data we collect, we explain it [here](#) in detail.

Question 1: Did you successfully login to twitter.com?

Yes ☐ No ☐

Question 2: How strong/secure do you think the password is that you just have entered on this website?

 N/A

Question 3: Do you agree with these statements?

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	N/A
The current website handles privacy sensitive information.	☐	☐	☐	☐	☐	☐
If someone steals your password for this website, they can harm you (e.g., financially, social reputation, use services, etc).	☐	☐	☐	☐	☐	☐

Figure 3: In situ questionnaire upon login to a new website.

Study participants from Mechanical Turk

	Survey	Invited to study	Participated
Number of participants	476	364	170
Gender			
Female	200	156 (78.0%)	73 (36.5%)
Male	274	208 (75.9%)	97 (35.4%)
Other	1	0	0
No answer	1	0	0
Age group			
18–30	180	139 (77.2%)	64 (35.6%)
31–40	178	135 (75.8%)	63 (35.4%)
41–50	71	58 (81.7%)	32 (45.1%)
51–60	35	24 (68.6%)	8 (22.9%)
61–70	11	7 (63.6%)	2 (18.2%)
≥71	1	1 (100%)	1 (100%)
Computer science background			
Yes	91	64 (70.3%)	27 (29.7%)
No	385	300 (77.9%)	143 (37.1%)
Native language			
English	468	358 (76.5%)	167 (35.7%)
Other	8	6 (75.0%)	3 (37.5%)

	Survey	Invited to study	Participated
Number of participants	476	364	170
Education level			
Less than high school	3	3 (100%)	1 (33.3%)
High school graduate	68	53 (77.9%)	26 (38.2%)
Some college, no degree	117	85 (72.6%)	34 (29.1%)
Associate's degree	79	64 (81.0%)	34 (43.0%)
Bachelor degree	174	133 (76.4%)	62 (35.6%)
Ph.D	2	1 (50.0%)	1 (50.0%)
Graduate/prof. degree	32	25 (78.1%)	12 (37.5%)
Other	1	0	0
Ethnicity			
White/Caucasian	355	274 (77.2%)	123 (34.6%)
Black/African American	50	38 (76.0%)	25 (50.0%)
Asian	31	23 (74.2%)	9 (29.0%)
Hispanic/Latino	27	21 (77.8%)	12 (44.4%)
Native American/Alaska	1	0	0
Multiracial	7	5 (71.4%)	1 (14.3%)
Other	5	3 (60.0%)	0

Summary statistics

Statistic	Mean	Median	SD	Min	Max
No. of passwords	10.39	9.00	5.52	1.00	27.00
Entry methods	2.24	2.00	0.75	1.00	4.00
Percentage reused passwords					
Non-reused	29.44%	21.58%	28.25%	0.00%	100%
Only-exact	15.72%	0.00%	24.43%	0.00%	100%
Only-partially	18.38%	11.11%	19.88%	0.00%	100%
Exact-and-partial	36.46%	38.75%	30.88%	0.00%	100%
Password composition					
Length	9.61	9.29	1.72	6.33	16.86
Character classes	2.52	2.50	0.58	1.00	3.94
Digits	2.54	2.38	1.24	0.25	6.73
Uppercase letters	0.85	0.67	0.81	0.00	4.62
Lowercase letters	5.92	5.72	1.96	1.67	15.50
Special chars	0.30	0.10	0.54	0.00	5.19
Password strength					
Zxcvbn score	2.20	2.14	0.75	0.67	4.00
Shannon entropy	29.31	28.37	7.93	16.00	68.00
NIST entropy	23.50	23.00	2.98	17.17	35.69

Entry method	All passwords	Unique passwords
Chrome auto-fill	949 (53.71%)	540 (51.67%)
Human	590 (33.39%)	331 (31.67%)
LastPass plugin	128 (7.24%)	100 (9.57%)
Copy&paste	55 (3.11%)	51 (4.88%)
Unknown plugin	41 (2.32%)	23 (2.20%)
External manager	4 (0.23%)	0 (0.00%)
Σ	1,767	1,045

Table 5: No. of password entries with each entry method.

By participant group: Strength and reuse

45
ppl

121
ppl

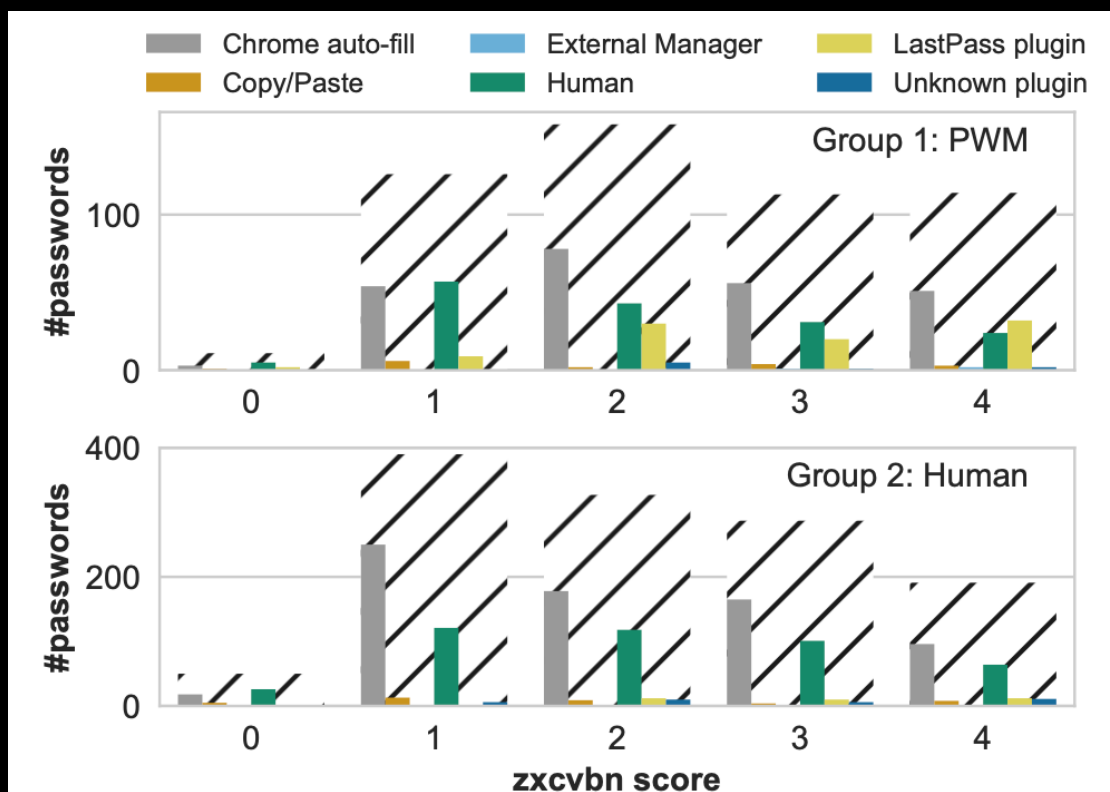


Figure 6: Password strength distribution by participant group and broken down by entry method. Hatched bars show total number of passwords per score. (Note the different y-axis limits)

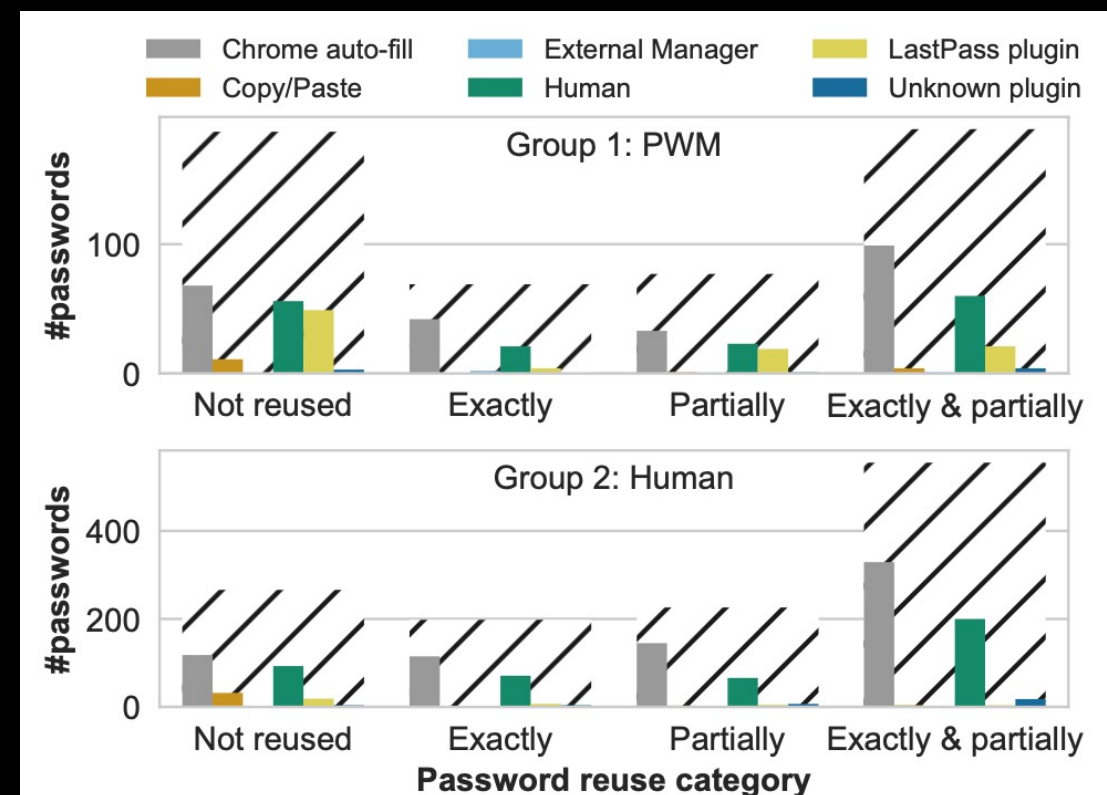


Figure 7: Distribution of reuse categories by participant group, broken down by entry method. Hatched bars show total number of passwords per category. (Note the different y-axis limits)

Modeling strength

	Estimate	Std. Error	z value	Pr(> z)
em:chrome	0.07	0.12	0.59	0.56
em:copy/paste	-0.13	0.35	-0.89	0.37
em:lastpass	0.24	0.35	0.69	0.49
em:unknownplugin	1.02	0.34	2.97	<0.01
in-situ:value	0.02	0.05	0.48	0.63
in-situ:strength	0.89	0.07	12.68	<0.001
user:entries	0.02	0.02	0.69	0.49
q9:generator	-0.45	0.67	-0.68	0.50
q14:memorize	-0.24	0.30	-0.79	0.43
q14:analog	0.05	0.29	0.16	0.88
q14:digital	0.09	0.31	0.29	0.77
q14:pwm	-0.16	0.28	-0.57	0.57
em:chrome * q9:gen.	2.30	0.60	3.84	<0.001
em:copy/paste * q9:gen.	3.40	1.22	2.79	<0.01
em:lastpass * q9:gen.	1.83	0.82	2.24	<0.05
em:unknownplugin * q9:gen.	0.22	1.34	0.16	0.87

em: Entry method; q9: Creation strategy; q14: Storage strategy; in-situ: Plugin questionnaire

Ordinal

Table 7: ~~Logistic~~ multi-level regression model predicting zxcvbn score. Estimates are in relation to manually entered passwords by a human. Statistically significant predictors are shaded. Interactions are marked with *.

- Creation strategy + EM significant together, but not individually
 - So: using such a password management tool only leads to significant improvement in the password strength when users also employ some supporting techniques (password generator) for the creation of their passwords
- Self-reported password strength was a significant predictor of the measured password strength

Modeling reuse

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.62	0.45	5.80	<0.001
em:chrome	0.46	0.16	2.81	<0.01
em:copy/paste	-2.68	0.41	-6.54	<0.001
em:lastpass	-1.05	0.37	-2.86	<0.01
em:unknownplugin	0.76	0.51	1.51	0.13
in-situ:value	-0.13	0.06	-2.01	<0.05
in-situ:strength	-0.21	0.08	-2.50	<0.05
user:entries	0.06	0.02	2.67	<0.01
q9:generator	-1.31	0.40	-3.24	<0.01
q14:memorize	0.22	0.25	0.88	0.38
q14:analog	-0.48	0.24	-1.98	<0.05
q14:digital	-0.18	0.26	-0.70	0.48
q14:pwm	-0.07	0.24	-0.30	0.76

em: Entry method; q9: Creation strategy; q14: Storage strategy; in-situ: Plugin questionnaire

Table 8: Logistic multi-level regression model predicting reuse. Estimates are in relation to manually entered passwords by a human and refer to the corresponding logit transformed odds ratios. Statistically significant predictors are shaded.

- Reuse was significantly influenced by the entry method of the password
 - Odds for reuse were 2.85 times *lower* by LastPass, 14.29 times lower if C&P
 - odds for reuse were 1.65 times *higher* by Chrome auto-fill
- Creation by alg: odds of non-reuse 3.70 times higher
- More passwords → greater odds of reuse
- Higher-value website → lower odds of reuse

Bottom line

- Users that rely on **technical support for password creation** had both **stronger and more unique passwords**
 - Even if entered through other channels than a manager
- **Chrome's auto-fill option increased password reuse** – more than 80% of Chrome auto-filled passwords were reused
 - Chrome at the time did not have password generation enabled by default

Discussion points

- Do you relate to the usage profiles given in the paper?
 - Partial or full reuse of passwords?
 - Different creation strategies and entry strategies?
- What has changed since 2018 as far as password managers go, which might affect the results?
- Do you have doubts about the validity or generalizability of some of these results? Why?